

BEREKET, Turkmenistan

WMO: 386470

Lat: 39.250N

Lon: 55.517E

Elev: 102

StdP: 14.64

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.5	21.4	2.4	6.2	31.7	7.4	8.1	34.7	30.3	38.0	27.2	32.9	14.3	90	0.511

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	106.1	68.3	103.4	68.2	101.0	68.2	76.2	94.5	74.8	93.1	73.6	91.7	8.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.6	113.1	86.9	68.7	105.9	86.9	66.8	98.9	87.2	39.7	94.2	38.4	93.2	37.2	91.5	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.0	23.3	20.6	13.4	111.5	9.8	2.9	6.3	113.6	0.6	115.3	-4.9	117.0	-12.0	119.1
			11.8	78.5	9.5	2.0	5.0	79.9	-0.6	81.1	-5.9	82.2	-12.8	83.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.7	37.8	40.5	51.6	63.5	76.1	85.4	89.9	88.3	77.4	63.3	49.1	40.1	(6)
	DBStd	20.08	8.23	9.38	7.85	8.29	6.12	5.30	4.32	4.71	6.15	7.95	8.58	7.38	(7)
	HDD50	1172	381	278	71	6	0	0	0	0	0	7	114	315	(8)
	HDD65	3460	842	687	421	124	5	0	0	0	3	126	480	772	(9)
	CDD50	6179	4	11	122	410	808	1061	1236	1187	823	421	87	9	(10)
	CDD65	2990	0	0	6	79	347	611	771	722	376	75	3	0	(11)
	CDH74	40700	0	0	115	810	3718	8595	11836	10845	4102	648	31	0	(12)
	CDH80	23712	0	0	33	289	1718	5109	7594	6846	1934	181	8	0	(13)
(14) Wind	WSAvg	8.3	9.2	9.6	8.7	8.5	7.2	7.7	7.5	7.3	8.6	8.0	8.5	9.4	(14)
(15) Precipitation	PrecAvg	6.7	0.8	0.8	1.1	0.9	0.6	0.1	0.2	0.1	0.1	0.6	0.7	1.0	(15)
	PrecMax	12.2	1.9	2.6	3.0	2.4	2.9	0.5	1.1	1.4	0.6	3.2	1.5	2.9	(16)
	PrecMin	3.0	0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
	PrecStd	2.1	0.4	0.7	0.7	0.6	0.6	0.2	0.4	0.3	0.1	0.7	0.5	0.8	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	64.2	68.6	84.6	93.6	100.9	107.1	109.8	109.8	100.7	90.3	77.9	67.5	(19)
		MCWB	49.6	51.8	58.3	63.1	65.0	68.0	69.6	68.4	64.5	63.5	58.3	51.1	(20)
	2%	DB	56.5	61.0	76.4	87.3	95.9	103.7	106.3	105.7	96.0	84.7	69.6	59.2	(21)
		MCWB	46.8	49.2	55.7	61.8	64.5	67.8	69.2	67.5	65.1	61.4	55.5	49.3	(22)
	5%	DB	51.6	56.4	70.0	82.7	92.4	100.9	103.6	102.6	93.0	80.9	65.4	54.3	(23)
		MCWB	44.9	47.1	54.0	60.1	64.1	67.2	69.0	67.9	64.7	60.1	53.3	47.0	(24)
	10%	DB	48.1	52.6	64.8	78.2	89.0	98.0	101.0	100.0	89.8	77.1	61.4	50.7	(25)
		MCWB	42.8	45.3	51.8	58.5	63.2	67.1	69.5	67.9	63.9	58.5	52.0	45.2	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.0	53.6	60.6	66.3	69.6	76.5	78.1	77.9	73.9	67.7	59.9	52.9	(27)
		MCDB	59.8	64.2	78.3	86.2	89.7	98.1	97.2	95.6	89.6	82.7	72.9	62.8	(28)
	2%	WB	47.7	50.3	57.2	63.7	67.9	73.3	76.3	76.1	71.4	65.2	57.2	49.9	(29)
		MCDB	55.1	58.9	71.2	82.9	87.0	94.1	94.4	94.2	87.6	79.4	67.4	57.6	(30)
	5%	WB	45.3	47.8	54.8	61.9	66.6	71.4	75.1	74.7	69.3	62.9	54.9	47.6	(31)
		MCDB	50.9	54.9	68.1	79.1	85.1	92.8	93.5	92.7	86.1	75.7	63.6	53.2	(32)
	10%	WB	43.2	45.8	52.5	60.0	65.5	69.7	74.0	73.3	67.0	60.7	52.9	45.6	(33)
		MCDB	47.8	51.8	63.4	75.0	83.5	89.9	92.2	90.9	85.0	73.3	60.2	50.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	15.2	18.6	19.9	20.8	21.4	20.0	21.6	21.9	21.0	16.3	13.4	(35)
		MCDBR	21.1	23.5	28.3	27.6	25.8	25.4	23.9	25.4	25.6	25.7	22.3	21.3	(36)
		MCWBR	13.0	13.5	12.7	10.6	8.6	9.3	8.6	9.7	11.0	11.7	11.7	13.4	(37)
	5% WB	MCDBR	19.2	21.0	24.9	24.8	21.4	21.0	19.0	18.3	20.2	21.0	19.3	18.4	(38)
		MCWBR	12.4	12.6	12.1	10.3	8.0	8.8	8.0	9.1	11.0	10.8	10.9	12.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.410	0.461	0.509	0.579	0.559	0.560	0.584	0.539	0.518	0.501	0.443	0.416	(40)
	taud		2.027	1.897	1.816	1.710	1.785	1.793	1.753	1.833	1.849	1.870	1.985	2.011	(41)
	Ebn at Noon		226	231	234	227	236	236	228	235	229	214	211	211	(42)
	Edh at Noon		39	51	62	73	69	69	71	64	59	52	41	37	(43)
(44) All-Sky Solar Radiation	RadAvg		681	946	1327	1681	2129	2429	2321	2163	1777	1234	832	610	(44)
	RadStd		65	106	89	173	101	71	93	105	63	67	61	62	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page